

Dear Editor, Angela Wulff,

We are pleased to resubmit our manuscript AMBI-D-26-00315 "The biodiversity we ignore: a global meta-analysis on taxonomic bias" for reconsideration at *AMBIO*. We thank you for the opportunity to revise and for the reviewers' thoughtful feedback.

We have carefully addressed all reviewer comments and believe the manuscript is substantially improved. Most suggestions were constructive and readily implementable, resulting in enhanced clarity. However, we feel it important to address upfront what we perceive as a fundamental methodological concern raised by Reviewer 2, as it speaks to the core contribution of our work.

On novelty and the value of quantitative synthesis:

Reviewer 2 questioned the novelty of our work, stating that "*the presence of the pattern the authors are looking at is known to us.*" We respectfully disagree with this assessment. While individual studies have documented taxonomic bias in specific contexts, no prior work has systematically quantified its magnitude, assessed its consistency across taxa and regions, or identified its drivers at a global scale using standardized effect sizes. This is precisely the contribution of meta-analysis: transforming scattered observations into robust quantitative patterns with sufficient statistical power to guide evidence-based interventions.

The claim that this pattern is "known" reflects what cognitive scientists call the curse of knowledge. Known to whom? Biodiversity bias experts? Ecologists broadly? Conservation practitioners? Policymakers? Funders? Students? Intuitive awareness among specialists does not equate to rigorous quantification, broad recognition across disciplines, or an evidence base sufficient to guide policy and funding decisions. Our synthesis provides exactly that foundation. As a meta-analysis, our novelty lies not in posing an unprecedented research question, but in providing the first comprehensive, quantitative answer to a long-standing one—a contribution model standard across scientific synthesis.

On heterogeneity and methodological rigor:

Reviewer 2 also expressed concern about heterogeneity in the included studies, noting that categorizing diverse studies involves unavoidable subjectivity and questioning whether averaging diverse measures is desirable. We acknowledge this concern and have provided more detail in the methods. However, we respectfully note that this concern reflects a fundamental misunderstanding of meta-analytic methodology.

Averaging across heterogeneous studies is not only standard but essential when heterogeneity is substantively meaningful—as it is in our case. Variation across studies reflects genuine differences in how bias manifests across contexts, taxa, and study designs. Our multilevel meta-regression framework was specifically designed to model this heterogeneity and identify its sources, not eliminate it. This approach

aligns with established best practices in ecological meta-analysis (Koricheva et al., 2013; Nakagawa & Santos, 2012). The alternative—restricting analysis to methodologically identical studies—would sacrifice external validity for artificial homogeneity and, in this case, leave no studies to analyze.

The reviewer acknowledges this is "effectively an unsolvable issue." We agree that synthesizing diverse literatures involves interpretive decisions, but this is true of all meta-analyses in ecology and evolution.

Our revised submission:

In response to reviewer feedback, we have:

- Enhanced the novelty statement in the Introduction to clarify our contribution
- Created a non-technical methods summary to improve accessibility (as suggested by reviewer 1)
- Created a section on Study Limitations section to explicitly address categorization subjectivity and other potential concerns
- Improved clarity throughout based on line-by-line suggestions from both reviewers

We have marked main changes to the text in orange.

We believe this revision addressed all technical concerns raised by reviewers while maintaining the methodological approach that makes this synthesis valuable. We hope that *AMBIO*, which has a strong tradition of publishing synthetic and policy-relevant biodiversity research, will find this contribution aligns with the journal's mission.

Thank you for your continued consideration of our work.

Sincerely,

Stefano Mammola

[On behalf of all co-authors]